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RESEARCH ARTICLE

4SWLR: A Switched System and Skid Steer Integrated Whole-Body Control Framework for Wheeled-Legged Robots

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ABSTRACT

Inspired by mammalian locomotion and vehicle skid-steering principles, this paper proposes a real-time motion planning and tracking control framework for wheeled-legged robots, integrating the obstacle-crossing capability of legged robots with the skid-steering mechanism of wheeled platforms. Unlike conventional wheeled-legged robot control methods that rely on external swing joints, the proposed framework leverages differential wheel actuation while comprehensively accounting for the robot-environment coupling effects under high-speed conditions, enabling efficient and stable high-speed steering. First, a hierarchical wheel-terrain contact dynamics model and a skid-steering kinematics model are established for wheeled-legged robots with skid-steering. By combining switched-system skid-steering kinematics with refined wheel–environment interaction dynamics, the framework effectively addresses active wheel torque control during high-speed steering. Second, a skid-steering-based motion paradigm is introduced, which co-optimizes legged dynamics and wheeled skid-steering kinematics, eliminating the need for continuous leg-lifting maneuvers to generate lateral forces and ensuring smooth high-speed steering. Finally, extensive experiments conducted in challenging environments—including staircases, trenches, ramps, single-side bridges, and unpaved terrains—validate the robustness and efficacy of the proposed approach. Comparative studies with state-of-the-art wheeled-legged control methods further demonstrate the superior mobility performance and enhanced wheel–terrain interaction dynamics achieved by our framework.

1 | Introduction

The exponentially growing demands in urban logistics and infrastructure maintenance have created an urgent need for mobile robotic systems capable of operating in complex, unstructured environments. Particularly in aging urban infrastructures and active construction sites, robotic platforms must demonstrate robust mobility and advanced obstacle negotiation capabilities when encountering diverse terrain features including staircases, drainage trenches, and uneven surfaces (He et al. 2022a; De Luca et al. 2023). This operational paradigm presents a fundamental engineering trade-off: robotic systems must simultaneously maintain energy efficiency, payload capacity, and terrain adaptability.

The biomechanical applications of quadrupedal robots represent a perfect fusion of biomechanics and robotic technology. By emulating the locomotion mechanisms of mammals (e.g., cats, cheetahs), they achieve adaptive mobility in complex terrains. However, their relatively low locomotion efficiency and limited payload capacity often fail to meet practical operational requirements. The wheel, recognized as one of humanity's

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greatest technological breakthroughs, has revolutionized transportation and significantly enhanced efficiency, becoming the cornerstone of modern transport and machinery. Yet wheeled vehicles depend on flat surfaces (Li et al. 2024), often losing mobility in complex terrains (e.g., trenches, stairs, or muddy ground) and lacking active obstacle-crossing capabilities, making them unsuitable for unstructured environments. To address these challenges, this paper integrates legged robots with wheeled vehicles, proposing an innovative motion control framework for skid-steering wheeled-legged robots that uniquely combines the maneuverability of skid-steering mechanisms with the obstacle-crossing capabilities of quadrupedal locomotion (Figure 1). The proposed solution demonstrates significant advantages in overcoming the mobility constraints characteristic of complex urban environments.

1.1 | Related Work

Wheeled-legged multimodal platforms have demonstrated superior performance in terms of speed, mobility, and transport efficiency compared to single-mode ground robots. One of the primary research focuses in wheeled-legged robotics is the integration of these two modes to achieve high-dynamics hybrid motion control (Medeiros et al. 2020).

Early research on wheeled-legged vehicles often compared the legs to the active suspension of traditional vehicles, with the joints acting as actuators to control parameters such as wheelbase and center-of-mass height (Qin et al. 2025). The HRC-HUBO (Lim et al. 2017) robot achieves functionality switching between wheeled and legged modes, where the robot curls up in wheeled mode and the legs remain stationary. The Momaro (Klamt and Behnke 2017) robot uses kinematic methods to drive the platform and overcome obstacles like stairs. The Centaur (Dadiotis et al. 2023; De Luca et al. 2022) robot adds steerable wheels to the feet, allowing omnidirectional motion and slow static actions, which were demonstrated in stepping stones, stairs, and initial attempts at stair climbing.

The field has witnessed remarkable progress in dynamic modeling, particularly through the integrated kinematic-dynamic bicycle model proposed in (Bellegarda and Nguyen 2021). This approach employs a nonlinear model predictive control framework that overcomes traditional limitations in drift motion planning while effectively addressing model ill-conditioning at low-speed regimes. Literature (Aguilera-Marinovic et al. 2017) introduces a generalized dynamic model for skid-steering mobile manipulators (SSMMs). This model innovatively incorporates a 6-DOF floating-base framework while accounting for transient wheel–terrain interaction effects. Literature (Zhu et al. 2025a) proposes a state-observer-based model predictive control (MPC) framework. By reformulating the nonlinear dynamics into a linear model with time-varying disturbances, the approach demonstrates robustness against payload variations and modeling uncertainties, as experimentally validated on the bipedal robot BRAVER.

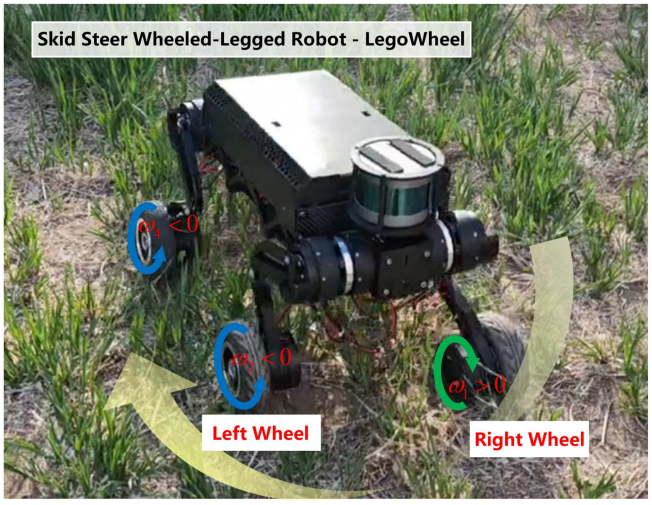


FIGURE 1 | The proposed skid-steering wheeled-legged robot platform.

Some researchers have added passive wheels to quadruped robots to achieve a wheel-skid effect, resembling a form of locomotion for four-legged robots. The passive-wheel equipped quadruped Anymal (Bjelonic et al. 2018) robot can perform a skating gait and glide over flat and inclined terrains using force control. The TAWL robot achieves a 50% reduction in slip rate on unstructured hard terrain through a kinematics-based control strategy (He et al. 2022b). The QSkater-E platform (Chen et al. 2024, 2023) achieves unique skating motions through its four-degree-of-freedom leg design that decouples foot positioning from passive wheel yaw control. The ANYmal system (Bjelonic et al. 2019, 2022) represents another milestone, employing active torque-controlled wheels modeled as dynamic contact points. Its control architecture combines offline MPC trajectory planning with real-time inverse dynamics tracking, enabling complex user-defined maneuvers. However, (Bjelonic et al. 2022) reports limitations during high-speed steering where joint actuators reached torque limits, suggesting the need for alternative steering mechanisms.

In recent advancements, the integration of Reinforcement Learning (RL) for motion control in wheeled-legged robots has shown significant potential (Vollenweider et al. 2023). Robots such as Lynx (Robotics 2024) and Unitree B2W (UNITREE 2024) have demonstrated impressive dynamic stability in rugged terrains, thanks to RL-enhanced control strategies. However, current simulation environments used for Reinforcement Learning are primarily designed for simulating "indoor robots operating on smooth, structured, and controlled surfaces" (NVIDIA 2024). These environments fail to account for the combined slip produced by tires during motion, which cannot fully replicate the dynamics and real-world effects needed to simulate real-world driving and off-road conditions. As a result, in the various videos showcasing complex motions learned through RL, high-speed turning scenarios are rarely observed.

1.2 | Work Superiority

This study investigates motion control for skid-steering wheeled-legged robots in complex environments. While existing achievements provide important references, several critical research gaps remain to be addressed.

- **Limitations in Wheeled-Legged Robot Research:** First, current studies on hybrid wheeled-legged systems primarily focus on high-dynamic obstacle-crossing capabilities and

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indoor/outdoor demonstrations, while largely neglecting the crucial dynamics of wheeled locomotion (Bjelonic et al. 2020). The forces generated by tire slip and their significant impact on overall robot dynamics, though fundamentally important, remain underexplored in wheeled-legged platforms. The intensified robot–environment coupling effects under high-speed conditions necessitate developing new control frameworks for practical implementation. Second, heading control during high-speed steering is particularly critical–employing leg-lifting steering methods (Bjelonic et al. 2021) may cause wheel lift-off, severely compromising stability. Moreover, the inherent high center-of-mass characteristic of legged architectures demands extremely high torque output from external joint actuators during high-speed turns, imposing stringent requirements on joint motor drive capacity. The ANYmal platform exemplifies this limitation, where interactive Trajectory Optimization (TO) methods led to torque saturation during high-speed steering, resulting in multiple control failures (Bjelonic et al. 2022). These issues reveal inherent deficiencies in traditional wheeled-legged robots with external swing joints for high-speed steering scenarios.

- **Limitations in Skid-Steering Vehicle Research:** Current theories on dynamics modeling and control for wheeled/tracked platforms have established relatively complete theoretical systems (Martínez et al. 2005; Liao et al. 2018; Trivedi et al. 2024; Tan et al. 2024). However, existing studies predominantly rely on the core assumption of "all wheels maintaining continuous ground contact," failing to adequately address skid-steering dynamics when wheels are actively lifted during controlled maneuvers. Consequently, all modeling approaches assume full wheel-ground contact. Due to their hybrid mobility characteristics, wheeled-legged robots can actively adjust ground friction by leg lifting (e.g., lifting the left front leg during left turns induces body tilting, utilizing gravity components to generate centripetal force). This multi-physics coordination mechanism transcends traditional skid-steering frameworks, offering novel solutions for high-mobility steering in complex terrains. However, it also introduces unresolved challenges including contact constraint violations, dynamic center-of-mass shifts, and time-varying friction properties that traditional skid-steering theories cannot address for wheeled-legged applications.
- **To solve these problems:** First, this study incorporates wheel dynamics and wheel-terrain contact dynamics when formulating motion control optimization problems, enhancing robotic wheel dynamics, and environmental coupling characteristics. Second, we adopt a jointless wheeled-legged robot design that structurally avoids insufficient lateral force generation from external swing motors during high-speed steering. Then, to compensate for the absence of lateral forces from swing joint motors, the control framework integrates skid-steering mechanisms to provide necessary lateral forces. Finally, addressing traditional skid-steering limitations for wheeled-legged robots, we propose switched-system wheel-terrain contact dynamics models and switched-system skid-steering kinematic models.

1.3 | Contribution

This work makes three key technical contributions to wheeled-legged robot control:

First, this study establishes a hierarchical wheel–terrain interaction dynamics model and skid-steering kinematics model specifically designed for wheeled-legged robots with skid-steering mechanisms. This approach incorporates an integrated dynamic model combining wheel dynamics with refined tire–terrain interaction physics, simultaneously addressing both wheel dynamics and contact dynamics. The proposed method effectively compensates for the critical deficiency in conventional wheeled-legged control that typically neglects tire slip effects, thereby achieving more precise and efficient motion tracking.

Second, this paper presents a skid-steering-based locomotion paradigm that eliminates the requirement for continuous leg-lifting maneuvers to generate lateral forces. We develop a specialized whole-body control architecture for reduced degree of freedom skid-steering wheeled-legged robots, which optimally coordinates wheel velocities with single rigid body dynamics through skid steering kinematic constraint. This approach enables sophisticated maneuver execution without reliance on external swing joint.

Third, the experimental results demonstrate that the robot exhibits exceptional terrain adaptability and sophisticated task-execution capabilities even without external swing joints. Notably, it can execute sharp turns at high speeds while maintaining remarkable stability when ascending and descending steep slopes. The system also shows consistent performance across challenging off-road conditions and various hard-surface terrains. Moreover, the robot successfully navigates stair-like obstacles comparable to its leg length using pre-planned trajectories, descends staircases with minimal disturbance, and reliably crosses wheel-span-sized trenches—collectively validating its robust mobility performance in diverse extreme scenarios.

2 | Robot System

The prototype platform features a biomimetic "front-elbow, back-knee" leg configuration, ensuring structural symmetry, and balanced mass distribution. As shown in Figure 2, the robot has a total weight of approximately 22 kg, with thigh and shank lengths of 0.15 m each, and is equipped with 6-inch pneumatic tires. The system comprises four wheel-leg subsystems. To avoid insufficient torque of the external swing motor when turning, the external swing joint motor was eliminated, resulting in three degrees of freedom (DOF) per subsystem—hip joint, knee joint, and wheel rotation—totaling 12 DOF. The hip and knee joint motors are fixed to the chassis at the hip position. The hip joint motor drives thigh rotation via a flange, while the knee joint motor controls knee movement through a parallel four-bar linkage mechanism. The wheel motor is mounted on the inner side of the shank's distal end, with its output shaft directly driving the wheel. All motors employ full-torque control, providing a peak torque of 33.5 N·m and a continuous maximum torque of 6 N·m, meeting the design requirements of the wheel-legged platform.

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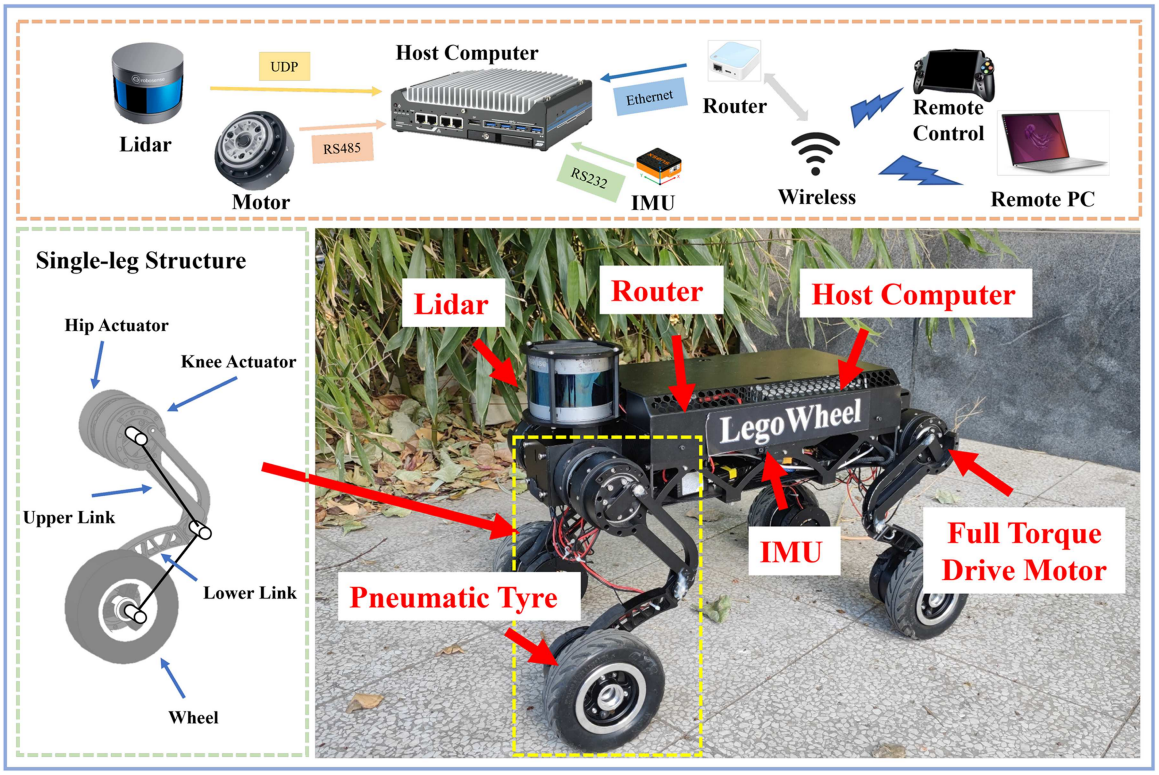


FIGURE 2 | Mechanical structure and electrical system of legowheel.

3 | Modeling

This paper presents a systematic investigation into the kinematic and dynamic modeling approaches for skid-steering wheel-legged robots, focusing on three critical aspects of their dynamic behavior: wheel-ground contact interactions, skid-steering kinematics, and wheel-leg system dynamics. The proposed modeling framework explicitly captures the hybrid nature of wheel-legged dynamics through a switched-system formulation, with particular emphasis on discrete contact state transitions during aggressive maneuvering operations. By integrating nonlinear tire force characteristics with multi-modal contact constraints, the developed model effectively bridges wheel-based dynamic properties and leg-oriented kinematic features. These integrated components collectively enable accurate prediction of coupled skid phenomena and contact force distribution, which are essential for achieving high-performance motion control across diverse terrain conditions.

3.1 | Switched-System Wheel-Ground Contact Model

The locomotion of wheeled-legged robots is fundamentally governed by wheel-ground interactions dominated by frictional forces. Conventional modeling approaches often neglect these critical interactions through simplified moving contact point assumptions in dynamic models, which significantly degrades control performance—particularly during high-speed, heavily loaded skid-steering maneuvers where large slip ratios occur. This oversight becomes especially detrimental when precise force prediction is required for dynamic stabilization and motion control.

This paper systematically evaluates multiple friction models (including the Magic Formula, brush model, and Dugoff model) to accurately capture these wheel–ground interactions. While the Magic Formula provides excellent accuracy, its computational intensity makes it less suitable for real-time optimization. The brush model and Dugoff model offer superior trade-offs between computational efficiency and modeling fidelity. The Dugoff model proves particularly advantageous for our application through: (1) its piecewise formulation enabling real-time computation, (2) explicit coupling of longitudinal/lateral force dynamics, and (3) accurate saturation characteristics under large slip conditions—all critical for modeling the combined slip phenomena during aggressive maneuvers. Unlike linear models that overestimate forces at high slip ratios, the Dugoff model's nonlinear representation of force coupling and saturation effects provides essential physical fidelity for dynamic control. Building upon this foundation, the present study establishes switched-system wheel–terrain contact dynamics constraints based on the Dugoff model to satisfy the contact characteristics of skid-steering wheeled-legged robots.

The Dugoff model calculates the longitudinal force F_i^x and lateral force F_i^y at the ith wheel as follows:

$$\begin{cases} F_i^x = \dfrac{c_x s_i}{1 - s_i} f(\kappa_i), \\ F_i^y = \dfrac{c_y \tan\alpha_i}{1 - s_i} f(\kappa_i), \end{cases} \tag{1}$$

where c_x and c_y are the longitudinal and lateral stiffness coefficients, respectively, $f(\kappa_i)$ is a piecewise function defining force

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saturation, α and s are obtained through the following expressions:

$$s_i = \begin{cases} \dfrac{w_i r_w - V_x + \dfrac{L_{12}}{2}\omega_z}{w_i r_w}, & i \in \{1, 3\}, \\ \dfrac{w_i r_w - V_x - \dfrac{L_{12}}{2}\omega_z}{w_i r_w}, & i \in \{2, 4\}. \end{cases} \tag{2}$$

$$\alpha_i = \begin{cases} \dfrac{V_y - \dfrac{L_{13}}{2}\omega_z}{V_x - \dfrac{L_{12}}{2}\omega_z}, & i \in \{1, 3\}, \\ \dfrac{V_y + \dfrac{L_{13}}{2}\omega_z}{V_x + \dfrac{L_{12}}{2}\omega_z}, & i \in \{2, 4\}. \end{cases} \tag{3}$$

In the above expression, ω_z is the yaw rate, V_x and V_y are the longitudinal and lateral velocities in the robot coordinate system, respectively; r_w is the wheel radius, L_{12} is the wheel track, and L_{13} is the wheelbase. $f(\kappa_i)$ is obtained through the following expressions:

$$f(\kappa_i) = \begin{cases} (2 - \kappa_i)\kappa_i, & \text{if } \kappa_i < 1, \\ 1, & \text{if } \kappa_i \geq 1. \end{cases} \tag{4}$$

The saturation threshold κ_i is given by:

$$\kappa_i = \frac{\mu F_i^z (1 - s_i)}{2\sqrt{(c_x s_i)^2 + (c_y \tan\alpha_i)^2}}, \tag{5}$$

where μ is the friction coefficient between tire and ground, F_i^z is the vertical load on the i-th wheel.

Adaptation of Dugoff Model for Legged Robots: While the Dugoff model was originally developed for automotive applications, its application to wheeled-legged robots requires several key adaptations: (1) The vertical load F_i^z varies dynamically due to leg lifting and body orientation changes, unlike vehicles where it remains relatively constant. (2) The slip ratio calculation must account for wheel lift-off events, where s_i becomes undefined. (3) The contact efficiency coefficients η_l and η_r compensate for partial wheel contact during leg-lifting maneuvers, which is unique to legged platforms. (4) The model must handle switched contact states, transitioning between full contact, partial contact, and no contact modes.

3.2 | Switched-System Skid Steering Kinematic Model

For skid-steering wheeled-legged robots, tire slip during turning maneuvers constitutes a significant factor that cannot be neglected. During motion, the system exhibits 16 distinct dynamic subsystems based on the contact states of individual wheels. This work establishes a comprehensive skid-steering kinematic model that accounts for wheel-ground contact conditions, grounded in Instantaneous Center of Rotation (ICR) theory (Martínez et al. 2005).

Based on rigid body motion theory, the following ICR coordinates are defined in the robot body-fixed coordinate system: the left wheel ICR $(x_{\text{ICRl}}, y_{\text{ICRl}})$, the right wheel ICR $(x_{\text{ICRr}}, y_{\text{ICRr}})$, and the robot ICR $(x_{\text{ICRv}}, y_{\text{ICRv}})$, as illustrated in Figure 3.

Through rigid body kinematics and superposition of wheel linear velocities, the lateral ICR coordinates are derived as:

$$y_{\text{ICRl}} = \frac{V_x - \eta_l \cdot \xi_l \cdot w_l r_w}{\omega_z}, \tag{6}$$

$$y_{\text{ICRr}} = \frac{V_x - \eta_r \cdot \xi_r \cdot w_r r_w}{\omega_z}, \tag{7}$$

where w_l and w_r denote the rotation velocities of left and right wheels, respectively, ξ_l and ξ_r represent correction factors for mechanical imperfections (e.g., tire underinflation) obtained through experimental measurement, and η_l and η_r are contact efficiency coefficients compensating for traction loss due to wheel lift-off.

The contact efficiency coefficients are calculated through Algorithm 1, which implements the following logic:

Algorithm 1 Contact Efficiency Coefficients Calculation.

Require: Wheel contact states σ_i where $\sigma_i = 1$ indicates contact and $\sigma_i = 0$ indicates lift-off, with $i = 1$: front left, $i = 2$: front right, $i = 3$: rear left, $i = 4$: rear right

Ensure: η_l, η_r (left/right contact coefficients)

{Left side contact efficiency calculation}
If $\sigma_1 = 1$ **and** $\sigma_3 = 1$ **then**
  $\eta_l \leftarrow 1.0$ {Both wheels grounded}
else if $\sigma_1 = 1$ **and** $\sigma_3 = 0$ **then**
  $\eta_l \leftarrow 0.7$ {Front grounded, rear lifted}
else if $\sigma_1 = 0$ **and** $\sigma_3 = 1$ **then**
  $\eta_l \leftarrow 0.5$ {Rear grounded, front lifted}
else
  $\eta_l \leftarrow 0.0$ {Complete lift-off}
end if
{Right side contact efficiency calculation}
If $\sigma_2 = 1$ **and** $\sigma_4 = 1$ **then**
  $\eta_r \leftarrow 1.0$ {Both wheels grounded}
else if $\sigma_2 = 1$ **and** $\sigma_4 = 0$ **then**
  $\eta_r \leftarrow 0.7$ {Front grounded, rear lifted}
else if $\sigma_2 = 0$ **and** $\sigma_4 = 1$ **then**
  $\eta_r \leftarrow 0.5$ {Rear grounded, front lifted}
else
  $\eta_r \leftarrow 0.0$ {Complete lift-off}
end if
return η_l, η_r

The contact efficiency coefficients η_l and η_r quantify the reduction in effective wheel-ground contact due to leg lifting.

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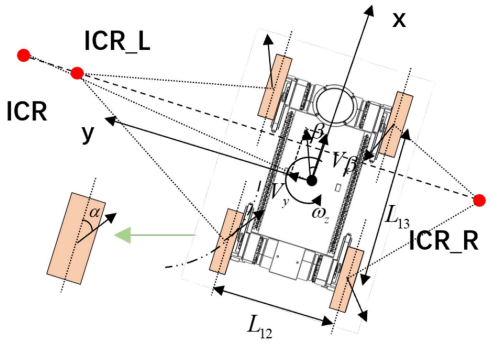


FIGURE 3 | Schematic of the kinematic model for a four-wheel skid-steering wheel-legged robot.

The coefficient values are determined as follows: (1) $\eta = 1.0$ when both wheels on a side are grounded, representing full contact efficiency. (2) $\eta = 0.7$ when the front wheel is grounded but the rear wheel is lifted, reflecting that approximately 70% of the normal contact force is maintained due to weight redistribution toward the front wheel. (3) $\eta = 0.5$ when the rear wheel is grounded but the front wheel is lifted, indicating lower efficiency as the rear wheel typically carries less load during forward motion. (4) $\eta = 0.0$ when both wheels are lifted, representing complete loss of contact.

These values were empirically determined through systematic simulation experiments following a rigorous methodology: (1) The robot model was simulated in a physics-based simulation environment with accurate contact force modeling. (2) The robot was tested under controlled simulation conditions with predefined leg lifting sequences, with 10–15 repetitions for each contact configuration to ensure statistical significance. (3) For each contact state, contact efficiency was calculated as $\eta = F_{\text{actual}}/F_{\text{theoretical}}$, where F_{actual} is the simulated traction force and $F_{\text{theoretical}}$ is the expected force assuming full contact (calculated from weight distribution and friction model). (4) The coefficient values represent mean values across all simulation trials, with standard deviations of approximately ± 0.05. (5) Validation was performed through independent simulation experiments including high-speed turning maneuvers with different leg lifting patterns and obstacle crossing scenarios requiring dynamic leg lifting, with ICR position predictions matching simulation measurements within 5% error. These simulation-determined values have been validated through experimental testing during various maneuvering scenarios and show consistent performance across different terrain types and operating conditions.

In the longitudinal direction, the fixed wheel-body connection yields identical ICR coordinates:

$$x_{\text{ICRl}} = x_{\text{ICRr}} = x_{\text{ICRv}} = -\frac{V_y}{\omega_z}. \tag{8}$$

The robot's yaw rate and longitudinal velocity can be reformulated as:

$$\omega_z = \frac{w_r r_w - w_l r_w}{y_{\text{ICRl}} - y_{\text{ICRr}}}, \tag{9}$$

$$V_x = \frac{y_{\text{ICRl}} \cdot w_r r_w - y_{\text{ICRr}} \cdot w_l r_w}{y_{\text{ICRl}} - y_{\text{ICRr}}}. \tag{10}$$

This leads to the compact matrix representation:

$$\begin{bmatrix} V_x \\ V_y \\ \omega_z \end{bmatrix} = \mathbf{A}_{\text{ICR}} \begin{bmatrix} w_l r_w \\ w_r r_w \end{bmatrix}, \tag{11}$$

where the ICR transformation matrix $\mathbf{A}_{\text{ICR}}$ captures the slip conditions and depends solely on wheel ICR coordinates:

$$\mathbf{A}_{\text{ICR}} = \begin{bmatrix} \frac{-y_{\text{ICRl}}}{y_{\text{ICRl}} - y_{\text{ICRr}}} & \frac{y_{\text{ICRl}}}{y_{\text{ICRl}} - y_{\text{ICRr}}} \\ \frac{-x_{\text{ICR}}}{y_{\text{ICRl}} - y_{\text{ICRr}}} & \frac{x_{\text{ICR}}}{y_{\text{ICRl}} - y_{\text{ICRr}}} \\ \frac{-1}{y_{\text{ICRl}} - y_{\text{ICRr}}} & \frac{1}{y_{\text{ICRl}} - y_{\text{ICRr}}} \end{bmatrix}, \tag{12}$$

3.3 | Wheel Dynamics Modeling

Traditional wheel-legged robot control methods often neglect wheel dynamics. This paper models the wheel dynamics and incorporates it through a Whole-Body Control (WBC) task formulation. The wheel dynamics are modeled considering both rotational inertia and tire–ground interaction forces. For each wheel i, the rotational dynamics can be expressed as:

$$I_w a_i = T_i - F_i^x r_w, \tag{13}$$

where a_i represents the wheel rotation accelerations, I_w represents the rotational inertia of the wheel, T_i represents the wheel torque. Equation (13) describes the longitudinal dynamics force analysis for a single wheel. The wheel force analysis diagram is shown in the bottom-right corner of Figure 4. The term F_i^x represents the longitudinal force in the wheel's forward direction, which is obtained from the contact force through rotational transformation. The term $F_i^x r_w$ represents the torque loss due to longitudinal friction.

4 | Motion Control

This study proposes a whole-body motion control framework for skid-steering wheel-legged robots (as shown in Figure 4), featuring a closed-loop architecture integrating MPC and WBC. The system supports multiple input interfaces, including but not limited to remote control commands and offline/online trajectory planning based on Nonlinear Programming (NLP) (Winkler et al. 2018). The MPC solver adopts a switched-system approach from reference (Farshidian 2022), dividing the system into 16 subsystems according to leg contact states, with gait transitions achieved through predefined mode sequences. For each subsystem, we establish switched wheel-ground contact models and skid-steering kinematic models, which are incorporated as constraints into the MPC system dynamics equations. The framework operates with MPC performing dual functions: optimizing desired trajectories based on current states while solving for desired inputs containing inverse

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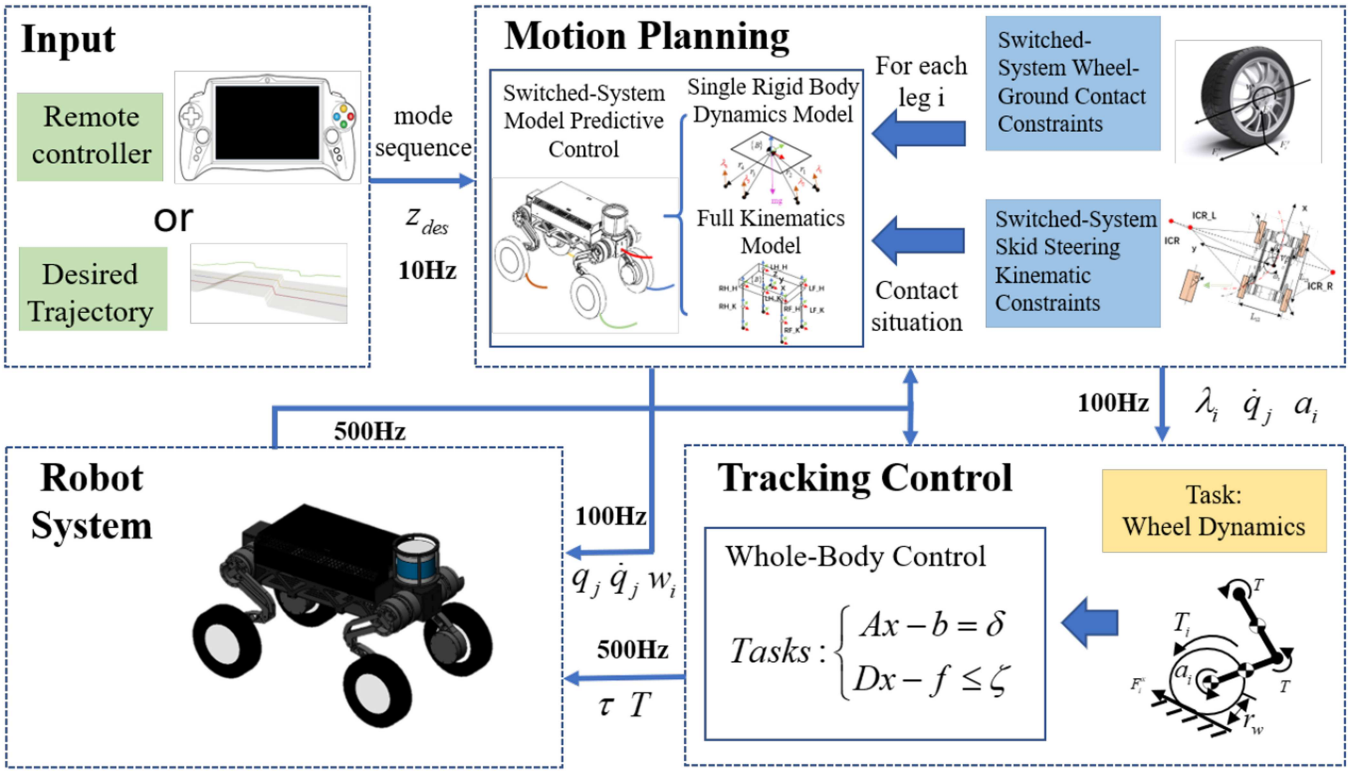


FIGURE 4 | A motion control framework for skid steer wheeled-legged robots.

dynamics parameters. Subsequently, the WBC computes actuation torques for joints and wheels using the optimized states and inputs from MPC.

MPC–WBC–CoM Architecture Relationship: The proposed control framework employs a hierarchical architecture: (1) The MPC layer operates at 100 Hz and optimizes trajectories for the Center of Mass (CoM) and wheel states over a prediction horizon, generating desired contact forces $\boldsymbol{\lambda}_i$ and wheel accelerations $\mathbf{a}_i$. (2) The WBC layer operates at 500 Hz and tracks the MPC outputs by solving inverse dynamics, computing joint torques $\boldsymbol{\tau}$ and wheel torques $\mathbf{T}$ that satisfy the full-body dynamics. (3) The CoM dynamics serve as the bridge: MPC uses simplified single rigid body dynamics centered at CoM for computational efficiency, while WBC uses full-body dynamics to ensure physical consistency. This two-layer approach enables real-time optimization while maintaining dynamic accuracy.

- **Actuation Strategy:**
  - **Joint Motors**
    - Control mode: Hybrid position-velocity-torque control
    - Position/velocity commands: MPC solutions
    - Torque commands: WBC solutions
  - **Wheel Motors**
    - Control mode: Hybrid velocity-torque control
    - Velocity commands: MPC solutions
    - Torque commands: WBC solutions

4.1 | Motion Planning

Unlike previous approaches, our steering mechanism employs skid steering and takes into account both wheel dynamics and wheel-ground contact. In the handling of tires, this paper integrates the skid steering kinematic model and the tire model as constraints within the MPC framework, and incorporate the wheel dynamics model as a constraint in the WBC task. Consequently, instead of using wheel position and velocity, this paper chooses the wheel speed and wheel acceleration as the wheel state variables and control inputs within the MPC. This approach allows for more precise modeling of wheel dynamics and wheel–ground interaction by adjusting the constraints, and enables contact-based skid steering.

4.1.1 | *System Model*

This paper employs a nonlinear switched-system MPC to address the nonlinearities and switching dynamics inherent in the motion of the wheeled-legged robot. The MPC receives offline trajectories generated by nonlinear programming (Winkler et al. 2018) or velocity and gait commands issued via a remote controller, and optimizes the motion trajectory in real-time. The system dynamics within the MPC are modeled using a rigid body dynamics model and a full kinematic model, with wheel kinematics and wheel-ground contact dynamics considered in both the state variables and constraints.

The single rigid body dynamics model is characterized by the centroidal state variables including the translational velocity $\mathbf{v}$, angular velocity $\boldsymbol{\omega}$, position $\mathbf{p}$, and orientation $\boldsymbol{\theta}$, with the contact forces $\boldsymbol{\lambda}_i^\top$ at the center of wheel (CoW) serving as control inputs, while the complete kinematic model requires augmenting the state space with joint positions $\mathbf{q}_j^\top$ and joint velocities $\dot{\mathbf{q}}_j^\top$ as additional states and inputs, respectively, and to account for skid-steering dynamics, the modeling framework further incorporates wheel rotation velocities $\mathbf{w}_i^\top$ as states and

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wheel rotation accelerations $\mathbf{a}_i^\top$ as inputs, where n and m represent the number of joints and wheels in the system. The state vector and control input vector are defined as follows:

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{v}^\top & \omega^\top & \mathbf{p}^\top & \theta^\top & \mathbf{q}_j^\top & \mathbf{w}_i^\top \end{bmatrix}^\top \in \mathbb{R}^{12+n_j+n_i}, \quad (14)$$

$$\mathbf{u}(t) = \begin{bmatrix} \lambda_i^\top & \dot{\mathbf{q}}_j^\top & \mathbf{a}_i^\top \end{bmatrix}^\top \in \mathbb{R}^{3n_e+n_j+n_i}. \quad (15)$$

where $n_j = 8$ is the number of joints, $n_i = 4$ is the number of wheels, and $n_e = 4$ is the number of legs. The dimensional consistency and mathematical operations of Equations (14) and (15) are clarified as follows:

State Vector Dimension: The dimension $\mathbb{R}^{12+n_j+n_i}$ is verified through component-wise analysis: (1) $\mathbf{v} \in \mathbb{R}^3$ (CoM translational velocity: longitudinal, lateral, vertical), (2) $\omega \in \mathbb{R}^3$ (CoM angular velocity: roll, pitch, yaw), (3) $\mathbf{p} \in \mathbb{R}^3$ (CoM position: x, y, z), (4) $\theta \in \mathbb{R}^3$ (CoM orientation: Euler angles), (5) $\mathbf{q}_j \in \mathbb{R}^{n_j}$ with $n_j = 8$ (joint positions), and (6) $\mathbf{w}_i \in \mathbb{R}^{n_i}$ with $n_i = 4$ (wheel rotation velocities). The total dimension is $3 + 3 + 3 + 3 + 8 + 4 = 24$, confirming $\mathbb{R}^{12+n_j+n_i} = \mathbb{R}^{24}$.

Input Vector Dimension: The dimension $\mathbb{R}^{3n_e+n_j+n_i}$ is verified as: (1) $\lambda_i \in \mathbb{R}^{3n_e}$ with $n_e = 4$ (contact forces at each leg's CoW: three components per leg × 4 legs = 12 dimensions), (2) $\dot{\mathbf{q}}_j \in \mathbb{R}^{n_j}$ with $n_j = 8$ (joint velocities), and (3) $\mathbf{a}_i \in \mathbb{R}^{n_i}$ with $n_i = 4$ (wheel rotation accelerations). The total dimension is $3 \times 4 + 8 + 4 = 24$, confirming $\mathbb{R}^{3n_e+n_j+n_i} = \mathbb{R}^{24}$.

4.1.2 | *Equations of Motion*

The system dynamics within the MPC are modeled using a rigid body dynamics model and a full kinematic model, with wheel kinematics and wheel-ground contact dynamics considered in both the state variables and constraints. The equations of motion used in this study are given as follows:

$$\begin{cases} \dot{\mathbf{v}} = \mathbf{g}(\theta) + \dfrac{1}{m}\displaystyle\sum_{i=1}^{4}\lambda_i, \\ \dot{\omega} = \mathbf{I}^{-1}(\mathbf{q})\Big(\dot{\mathbf{I}}(\mathbf{q},\mathbf{u}) - \omega \times \mathbf{I}(\mathbf{q})\omega + \displaystyle\sum_{i=1}^{4}\mathbf{r}_{fi}(\mathbf{q}) \\ \qquad \times \lambda_i\Big), \\ \dot{\mathbf{p}} = \mathbf{R}(\theta)\mathbf{v}, \\ \dot{\theta} = {}^{W}T_B(\theta)\Big(\omega - \mathbf{B}\mathbf{J}_{\text{com}}^{\omega}(\mathbf{q})\mathbf{u}\Big), \\ \dot{\mathbf{q}}_j = \mathbf{u}_j, \\ \dot{\mathbf{w}}_i = \mathbf{a}_i. \end{cases} \quad (16)$$

where $\mathbf{g}$ is the gravitational acceleration in the world coordinate system, m is the total mass of the rigid body, $\mathbf{I}$ is the moment of inertia about the center of mass, $\mathbf{R}$ is the rotation matrix representing the body's orientation relative to the world coordinate system, ${}^{W}T_B$ is a matrix which transforms the angular velocities in the base frame to the Euler angles derivatives in the global frame, and ${}_{B}J_{com}^{\omega}$ is the Jacobian matrix of CoM rotation with respect to robot base frame, $\mathbf{r}_{fi}$ represents the position of the CoW.

4.1.3 | *Constraints*

For switched systems MPC, constraints are typically designed separately for individual subsystems, as demonstrated in prior quadruped robot research which established zero contact force constraints for swing legs and friction cone constraints for stance legs (Farshidian et al. 2017); to address the high-speed and heavy-load motion characteristics of skid-steering wheeled-legged robots, this study innovatively introduces skid-steering kinematic constraints into the switched system to ensure the MPC optimization can effectively achieve steering through differential speeds, while incorporating wheel-terrain contact dynamics constraints to more accurately characterize wheel-terrain contact dynamics; beyond these four fundamental constraints, the formulation is further enhanced with a non-holonomic rolling constraint requiring zero velocity of contact wheels along the terrain's normal plane, thereby establishing a comprehensive five-constraint framework for the switched system MPC optimization.

4.2 | Tracking Control

Using MPC, reference trajectories for the center of mass, wheels, and joints, as well as contact forces and torques, are computed. Next, inverse dynamics is used to calculate the desired joint torques and track the planned trajectories. WBC follows the Hierarchical Optimization (HO) framework proposed in (Bellicoso et al. 2016). Within this framework, tracking control is formulated as a cascade of Quadratic Programming (QP) problems composed of linear equality and inequality constraints. Task is defined in the following form:

$$Tasks: \begin{cases} \mathbf{A}\mathbf{x} - \mathbf{b} = \delta, \\ \mathbf{D}\mathbf{x} - \mathbf{f} \le \zeta. \end{cases} \quad (17)$$

where the linear equality constraints are defined by $\mathbf{A}$ and $\mathbf{b}$, and the linear inequality constraints are defined by $\mathbf{D}$ and $\mathbf{f}$. δ and ζ represent slack variables to be minimized. The optimization solution vector $\mathbf{x}$ is constructed as:

$$\mathbf{x} = \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \\ \tau \\ \mathbf{T} \\ \mathbf{a} \end{bmatrix} \in \mathbb{R}^n, \quad (18)$$

where $n = 42$ is the dimensionality of $\mathbf{x}$, $\tau \in \mathbb{R}^{n_j}$ represents joint torques, $\mathbf{T}$ represents the wheel torque, $\ddot{\mathbf{q}} \in \mathbb{R}^{6+n_j}$ is the generalized acceleration vector, and the generalized position vector $\mathbf{q}$ is expressed as:

$$\mathbf{q} = \begin{bmatrix} \mathbf{p} \\ \theta \\ \mathbf{q}_j \end{bmatrix} \in \mathbb{R}^{6+n_j}. \quad (19)$$

4.2.1 | *Equations of Motion*

The motion equations of the wheel-legged robot can be expressed as (Bellicoso et al. 2016):

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$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{S}^{\top}\tau + \mathbf{J}_s^{\top}\lambda, \tag{20}$$

where $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{(6+n_j)\times(6+n_j)}$ is the mass matrix, $\mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{6+n_j}$ is the vector of Coriolis, centrifugal, and gravitational terms, $\mathbf{S} = [\mathbf{0}_{n_j\times 6} \quad \mathbf{I}_{n_j\times n_j}]$ is the selection matrix for actuated degrees of freedom, $\mathbf{J}_s^{\top} \in \mathbb{R}^{3\cdot \text{ncn_cnc}\times(6+n_j)}$ is the support Jacobian, and n_c represents the number of contact legs. This can be reformulated into equality constraint form as:

$$\begin{bmatrix}\mathbf{M} & -\mathbf{J}_s^{\top} & -\mathbf{S}^{\top} & \mathbf{0}_{(6+n_j)\times 2n_i}\end{bmatrix} \cdot \mathbf{x} = h(\mathbf{q}, \dot{\mathbf{q}}). \tag{21}$$

4.2.2 | *Task Space Motion Tracking*

Unlike quadrupeds, wheels generate motion regardless of whether they are in the swing or support phase. Therefore, the planned wheel velocities need to be tracked. The velocity of the CoW in the world frame can be computed as:

$$\mathbf{v}_{fi} = \mathbf{J}\dot{\mathbf{q}}, \tag{22}$$

where $\mathbf{J}$ is the Jacobian mapping leg joint velocities to operational space linear velocities of a wheel. Taking the derivative yields the CoW acceleration:

$$\dot{\mathbf{v}}_{fi} = \dot{\mathbf{J}}\dot{\mathbf{q}} + \mathbf{J}\ddot{\mathbf{q}}. \tag{23}$$

This leads to the following equality constraint:

$$\begin{aligned}[\mathbf{J} \quad \mathbf{0}]\mathbf{x} = K_D \cdot \left(\mathbf{v}_{\text{fi}}^{\text{des}} - \mathbf{v}_{\text{fi}}^{\text{mea}}\right) \\ + K_P \cdot \left(\mathbf{r}_{\text{fi}}^{\text{des}} - \mathbf{r}_{\text{fi}}^{\text{mea}}\right) - \dot{\mathbf{J}}\dot{\mathbf{q}}. ,\end{aligned} \tag{24}$$

where K_P and K_D are proportional and derivative gains, $\mathbf{r}_{fi}^{des}$ and $\mathbf{v}_{fi}^{des}$ are the desired position and velocity, and $\mathbf{r}_{fi}^{mea}$ and $\mathbf{v}_{fi}^{mea}$ are the measured position and velocity, respectively.

4.2.3 | *Contact Force/Wheel Acceleration*

The MPC-planned contact forces and wheel accelerations are tracked using the following constraints:

$$\begin{bmatrix}\mathbf{0}_{3n_e\times(6+n_j)} & \mathbf{I}_{3n_e\times 3n_e} & \mathbf{0}_{3n_e\times(n_j+2n_i)}\end{bmatrix}\mathbf{x} = \lambda_{\text{des}}, \tag{25}$$

$$\begin{bmatrix}\mathbf{0}_{n_i\times(6+2n_j+3n_e)} & \mathbf{I}_{n_i\times n_i} & \mathbf{0}_{n_i\times n_i}\end{bmatrix}\mathbf{x} = \mathbf{a}_{\text{des}}. \tag{26}$$

4.2.4 | *Wheel Dynamics*

The rotational dynamics of the wheels are written as an equality constraint:

$$\begin{bmatrix}\mathbf{0} & {}^{D}\mathbf{R}_W & \mathbf{0} & \frac{1}{r_w} * \mathbf{I}_{n_i\times n_i} & -\frac{I_w}{r_w} * \mathbf{I}_{n_i\times n_i}\end{bmatrix}\mathbf{x} = \mathbf{0}, \tag{27}$$

where ${}^{D}R_W$ is the matrix calculating the contact force component along the wheel's forward direction.

4.2.5 | *Torque Limits*

Joint and wheel torque limits are imposed through upper and lower bounds:

$$\begin{bmatrix}\mathbf{0}_{n_i\times(6+n_j+3n_e)} & \mathbf{I}_{n_j\times n_j} & \mathbf{0}_{2n_i\times 2n_i} \\ \mathbf{0}_{n_i\times(6+n_j+3n_e)} & -\mathbf{I}_{n_j\times n_j} & \mathbf{0}_{2n_i\times 2n_i}\end{bmatrix}\mathbf{x} \le \begin{bmatrix}\tau^{\max} \\ -\tau^{\min}\end{bmatrix}, \tag{28}$$

$$\begin{bmatrix}\mathbf{0}_{n_i\times(6+2n_j+3n_e)} & \mathbf{I}_{n_i\times n_i} & \mathbf{0}_{n_i\times n_i} \\ \mathbf{0}_{n_i\times(6+2n_j+3n_e)} & -\mathbf{I}_{n_i\times n_i} & \mathbf{0}_{n_i\times n_i}\end{bmatrix}\mathbf{x} \le \begin{bmatrix}T^{\max} \\ -T^{\min}\end{bmatrix}. \tag{29}$$

4.2.6 | *Contact Force Limits*

To approximate friction cone constraints on the terrain plane, WBC employs a QP formulation that simplifies the friction cone into a pyramid structure:

$$\begin{cases}-F_i^z \le 0 \\ F_i^x - \mu^* F_i^z \le 0 \\ -F_i^x - \mu^* F_i^z \le 0, \\ F_i^y - \mu^* F_i^z \le 0 \\ -F_i^y - \mu^* F_i^z \le 0\end{cases} \tag{30}$$

where μ is the ground friction coefficient.

4.3 | Constraint Parameter Determination

The constraint parameters in the optimization framework are determined through a combination of hardware specifications, experimental measurement, and analytical safety margins.

Joint and Wheel Torque Limits: The torque bounds $\tau^{\max}$, $\tau^{\min}$, $T^{\max}$, and $T^{\min}$ are derived from motor specifications: peak torque: 33.5 N·m, continuous maximum: 6 N·m. **Friction Cone Constraints:** The friction coefficient μ in Equation (30) is determined through experimental measurement using the deceleration method. The robot is driven at a certain initial velocity on a specific terrain surface and then brought to a complete stop through wheel lock-up via emergency braking.

5 | Results

To evaluate our motion controller, we conducted a series of quantitative experiments focusing on agile maneuvers using the same skid-steering wheel-legged robot, LegoWheel. Table 1 summarizes three motion controllers and compares their functional capabilities when deployed on LegoWheel. The subsequent sections present experimental results and analyses conducted with LegoWheel, with visual demonstrations available in the Supporting Video Movie S1.

5.1 | Experimental Setup

Its reliability and effectiveness have been rigorously validated through extensive experiments conducted in various challenging real-world scenarios. Both remote controller commands without perception and offline trajectories generated via nonlinear programming (Winkler et al. 2018) were successfully

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TABLE 1 | Performance comparison of motion controllers.

Metric	RL controller	MPC controller	Proposed
Skid-steering	√	×	√
Freq. (Hz)	100	100	100
Vel. tracking	Medium	Medium	Good
Yaw tracking	Poor	×	Good
Uneven terrain	√	√	√
Swing joints req.	×	√	×
High-speed rel.	Low	×	High

TABLE 2 | Task prioritization in wheel-legged robot control.

Priority	Type	Task
1	=	Equations of Motion
2	≥	Torque Limits
3	≥	Contact Force Limits
4	=	Wheel Dynamics
5	=	Task Space Motion Tracking
6	=	Contact Force
7	=	Wheel Acceleration

tracked, demonstrating the scalability of the proposed algorithm. The priority hierarchy of the WBC algorithm is presented in Table 2.

The robot is equipped with a 16-line LiDAR, a 9-axis IMU, 12 fully torque-controlled actuators, a compact router for communication, and an industrial PC powered by an Intel i9-14900 processor. The following algorithms were deployed on the same industrial PC:

- An MPC-based trajectory optimization algorithm running at 100 Hz,
- A WBC-based tracking control algorithm running at 500 Hz, and
- A state estimation algorithm (Zhu et al. 2025b) also running at 500 Hz.

The MPC optimization was solved using the open-source nonlinear switched-system solver OCS2 (Farshidian 2022), while the WBC optimization was solved using the open-source QP solver qpOASES (Ferreau et al. 2014). The kinematic and dynamic models (excluding the skid-steer kinematics, wheel dynamics, and tire models) were computed using the rigid body dynamics library Pinocchio (Carpentier et al. 2019).

5.2 | Control Parameter Selection and Tuning

The selection and tuning of control parameters is critical for achieving robust and high-performance control of wheeled-legged robots. This section provides a comprehensive guide on how control parameters are chosen and adjusted for different tasks, addressing both the MPC layer parameters including prediction horizon, control horizon, and cost function weights, and the WBC layer parameters including stiffness and damping gains. Additionally, the control frequency selection and its impact on system performance are analyzed in detail.

5.2.1 | *Control Frequency Selection*

The MPC control frequency of 100 Hz was selected through systematic evaluation of performance versus computational cost. To understand the impact of different control frequencies, comprehensive experiments were conducted at 25, 50, 100, and 200 Hz. **Low Frequency Analysis 25 Hz:** At 25 Hz, the control loop updates every 40 ms, which is insufficient for capturing rapid dynamic maneuvers. Experimental results showed that tracking errors increased by 25%–30% compared to 100 Hz, particularly during rapid steering maneuvers and terrain transitions. The longer update interval leads to delayed response to disturbances and reduced prediction accuracy, as the system state may change significantly between control updates. **Medium Frequency Analysis 50 Hz:** At 50 Hz with a 20 ms update interval, performance improved but still showed 15%–20% degradation in tracking accuracy compared to 100 Hz. The system exhibited noticeable lag during high-speed maneuvers, with yaw rate tracking errors increasing during rapid direction changes. **Optimal Frequency 100 Hz:** The 100 Hz frequency with a 10 ms update interval provides an optimal balance between prediction accuracy and computational feasibility. This frequency ensures: first, sufficient temporal resolution to capture dynamic maneuvers with characteristic time constants on the order of 50–100 ms; second, real-time optimization feasibility on the available hardware including the Intel i9-14900 processor, with average computation time per control cycle below 10 ms; and third, compatibility with the 500 Hz WBC layer through appropriate interpolation, maintaining smooth reference trajectories. **High Frequency Analysis 200 Hz:** At 200 Hz with a 5 ms update interval, marginal improvement of less than 0.5% in tracking accuracy was observed, but the computational burden increased significantly. The average computation time approached 8–9 ms per cycle, leaving insufficient margin for real-time execution and risking control loop delays. Additionally, the benefits of higher frequency diminish due to actuator bandwidth limitations and sensor update rates. The comparative performance metrics across different frequencies are summarized in Table 3,

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demonstrating that 100 Hz achieves the best trade-off between tracking accuracy and computational efficiency.

5.2.2 | *MPC Parameter Selection*

Prediction and Control Horizons: The MPC prediction horizon is set to 1.0 s with a discretization time step of 0.01 s (approximately 100 prediction steps). These values were selected through systematic evaluation: shorter horizons resulted in insufficient prediction accuracy for dynamic maneuvers, while longer horizons exceeded computational limits. The selected horizon captures two to three characteristic time constants of the system dynamics, balancing prediction accuracy and computational feasibility. **State Tracking Weight Matrix (Q Matrix):** The weights for each state component are listed in Table 4. The tuning procedure follows: (1) Initial weights set to equal values, (2) Increase weights for critical states (yaw rate: 300, longitudinal velocity: 500.0, position x: 5000) until tracking errors are minimized, (3) Adjust weights for less critical states to prevent excessive control effort. The priority hierarchy prioritizes yaw rate and longitudinal motion (highest weights), followed by position tracking, while vertical motion and roll/pitch have lower weights to allow natural body motion. **Input Penalty Weight Matrix (R Matrix):** The weights are listed in Table 4. Joint velocities have high penalty (80,000) to prevent excessive motion, while wheel accelerations have low penalty (0.01) to allow aggressive skid-steering control. Contact force weights balance stability (z-direction: 1) and lateral force prevention (x-, y-directions: 10). **Task-Specific Parameter Adjustment:** For high-speed tracking tasks, increase velocity tracking weights. For precision positioning tasks, increase position tracking weights. For energy-efficient operation, increase input penalty weights.

TABLE 3 | Performance comparison across different control frequencies.

Freq.	Yaw RMSE (rad/s)	Long. Vel. RMSE (m/s)	Comp. Time (ms)
25 Hz	0.082	0.045	2.1
50 Hz	0.065	0.032	4.8
100 Hz	0.056	0.021	7.2
200 Hz	0.053	0.020	8.9

TABLE 4 | Control parameters.

Parameter	Value
MPC prediction horizon	1.0 s
MPC discretization time step (dt)	0.01 s
State Tracking Weights (Q Matrix):	
CoM velocity $\mathbf{v}$ (longitudinal, lateral, vertical)	500.0, 500.0, 100.0
CoM angular velocity ω (roll, pitch, yaw)	10, 30, 300
CoM position $\mathbf{p}$ (x, y, z)	5000, 1500, 1500
CoM orientation θ (all axes)	300
Joint positions $\mathbf{q}_j$ (all joints)	500.0
Wheel velocities $\mathbf{w}_i$ (all wheels)	5000
Input Penalty Weights (R Matrix):	
Contact forces λ_i (x, y, z)	10, 10, 1
Joint velocities $\dot{\mathbf{q}}_j$ (all joints)	80000
Wheel accelerations $\mathbf{a}_i$ (all wheels)	0.01
WBC position K_P	350
WBC velocity K_D	37

5.2.3 | *WBC Gain Selection*

The proportional and derivative gains in Equation (24) are set as $K_P = 350$ for position tracking and $K_D = 37$ for velocity tracking (see Table 4). The tuning procedure: (1) Start with conservative gains based on system natural frequencies, (2) Gradually increase gains while monitoring tracking performance and stability, (3) Ensure bandwidth is 5–10 times the MPC update frequency without exciting high-frequency modes, (4) Validate robustness with ±20% parameter variations. The selected gains provide 70%–80% of critical damping, ensuring good transient response. Gains are applied at the task space level; for different robot configurations, scale proportionally to effective inertia.

5.2.4 | *Parameter Robustness and Guidelines*

To demonstrate robustness, the controller was tested with ±20% variations in all parameters. The system maintained stable performance, indicating reasonable margins for different operating conditions. For significantly different robot platforms or terrain types, parameter re-tuning may be necessary. **General Guidelines for Parameter Selection:** (1) Prediction horizon: cover two to three characteristic time constants of system dynamics. (2) State tracking weights (Q): reflect task priorities, typically 10–100 times larger than input penalty weights. Critical states (yaw rate, longitudinal velocity) should have higher weights. (3) Input penalty weights (R): balance control effort and responsiveness–high penalties for inputs to minimize (joint velocities), low penalties for aggressive control (wheel accelerations). (4) Control gains: provide 70%–80% of critical damping, with bandwidth 5–10 times the control frequency. (5) Control frequency: 10–20 times the system's natural frequency, limited by computational constraints. These guidelines, combined with the specific values in Table 4, provide a reproducible starting point for similar wheeled-legged robot platforms.

5.3 | Complex Unstructured Terrains

As shown in Figure 5, the proposed algorithms were tested on a variety of challenging terrains in both perception-based and non-perception-based motion modes:

1. **Ascending pre-planned stair trajectories.** The robot tracked a pre-planned trajectory to climb stairs with a height of **0.15 m**, which is equal to the length of the robot's legs. Due to insufficient leg length, both wheels needed to be lifted in order to reach the next step. Despite these constraints, the robot successfully climbed the stairs.
2. **Descending stairs without perception.** Without pre-planned motion trajectories, the robot was given a slow forward velocity. It adapted to the terrain incrementally and avoided toppling.

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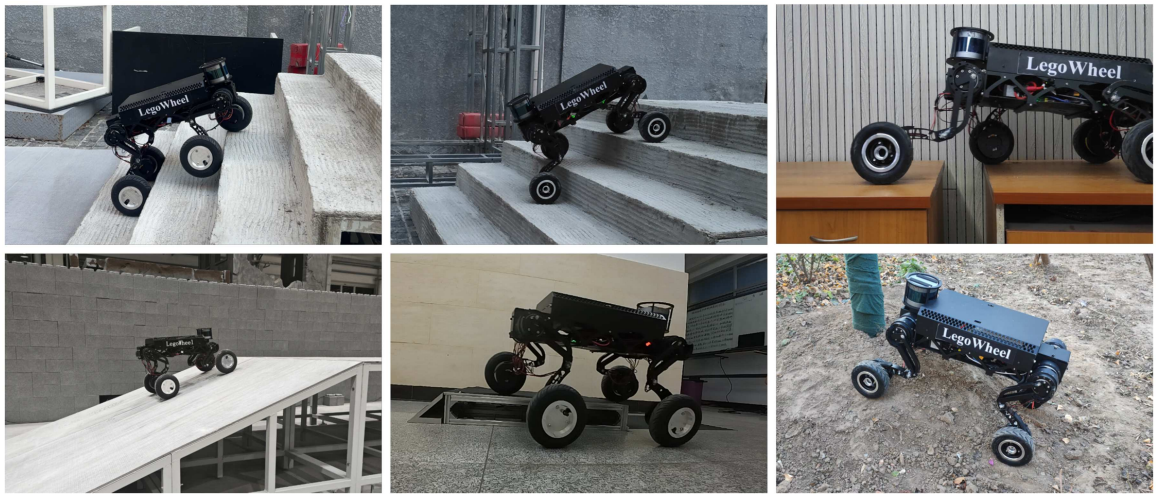


FIGURE 5 | The figure illustrates the testing of robot in various challenging environments.

3. **Crossing a 0.15 m trench.** The robot successfully tracked a pre-planned trajectory to avoid wheel entrapment in the trench. In contrast, the controller trained with perception-free reinforcement learning failed to accomplish this task (Winkler et al. 2018), highlighting the critical importance of contact force distribution and trajectory planning.
4. **Traversing a 25° slope without perception.** The robot gradually adjusted its posture to stabilize the body and maintain a level orientation while ascending and descending slopes.
5. **Crossing a 0.12 m narrow bridge without perception.** The robot stabilized its body while lifting one leg higher to traverse the bridge successfully.
6. **Outdoor terrain tests.** The robot was tested on multiple outdoor surfaces, including dirt paths, stone-paved walkways, and concrete roads, achieving stable motion at a speed of **2 m/s**.

5.3.1 | *Statistical Evaluation*

To ensure statistical significance, each experiment was conducted with 8–12 independent trials depending on the experiment type. Performance metrics including success rates and failure counts were recorded and analyzed. The results are reported in Table 5. A trial was considered successful if the robot completed the task without falling or requiring manual intervention. The success rate is calculated as the percentage of successful trials out of the total number of trials.

5.3.2 | *Failure Cases and Limitations*

During extensive testing, a failure case was encountered in one experiment type as reported in Table 5: **Obstacle crossing:** One failure occurred out of 10 trials (90.0% success rate), caused by insufficient contact force distribution during the transition phase when lifting wheels over obstacles. The failure occurred when the robot's center of mass shifted too rapidly during wheel lift-off, leading to an unstable configuration that could not be recovered. Despite this isolated failure, the overall success rates demonstrate the robustness of the proposed framework. The failure highlights the importance of: (1) improved contact force distribution during dynamic transitions, (2) more conservative safety margins for extreme operating conditions, and (3) enhanced state estimation accuracy during complex–terrain interactions.

TABLE 5 | Statistical evaluation results.

Experiment Type	Trials	Failures	Success (%)
Stair climbing	8	0	100
Descending stairs	10	0	100
Obstacle crossing	10	1	90.0
Traversing 25° slope	10	0	100
Crossing 0.12 m bridge	10	0	100
Uneven terrain	12	0	100

5.4 | Comparison With Model-Free Reinforcement Learning Methods

This paper systematically compared the performance of the proposed model-based controller with an advanced RL motion controller (Robotics 2024) on the same hardware platform across different terrains. For this comparison, ten independent experimental trials were conducted for each controller, and the median trial was selected for performance analysis to ensure robustness against outliers and environmental variations. As shown in Figure 6, both controllers were tested on asphalt pavement for 1 rad/s yaw rate tracking using the median trial data. The theoretical predictions in Figure 6 show good agreement with experimental measurements, validating the accuracy of the proposed dynamic models. The RL controller exhibited three critical limitations: (1) A root-mean-square tracking error of 78.04%, compared with 5.61% achieved by the proposed method, representing a reduction of 72.43 percentage points, due to simulation-to-reality gaps in wheel–terrain interaction modeling; (2) Frequent unintended leg lifting during steering,

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indicating misinterpretation of tire slip dynamics as collision events; (3) An oversimplified velocity control strategy that causes excessive tire slip during skid steering on high-friction surfaces, resulting in insufficient steering torque and response lag. These factors collectively reduce stability margins, with rollover risk increasing significantly at speeds exceeding 1 m/s.

In contrast, the proposed method explicitly considers wheel dynamics and tire-ground contact models, jointly optimizing motor speed and feedforward torque. The feedforward-feedback torque coordination mechanism further ensures performance consistency across different friction coefficients, achieving faster response, higher robustness, and superior tracking accuracy.

5.5 | Comparison With Simplified Contact Point Models

To quantify the advantages of wheel dynamics modeling, this paper benchmarked against a conventional MPC implementation that treats wheels as moving contact points (Bjelonic et al. 2022). For this comparison, ten independent experimental trials were conducted

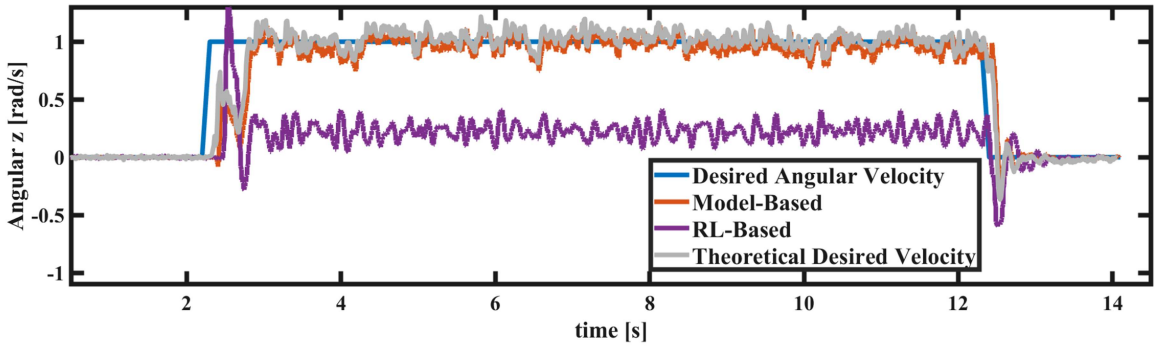


FIGURE 6 | Comparison of yaw rate tracking performance between reinforcement learning (purple) and the proposed model predictive control (orange-red) under 1 rad/s reference command. Gray solid lines denote curves constructed from theoretical equations.

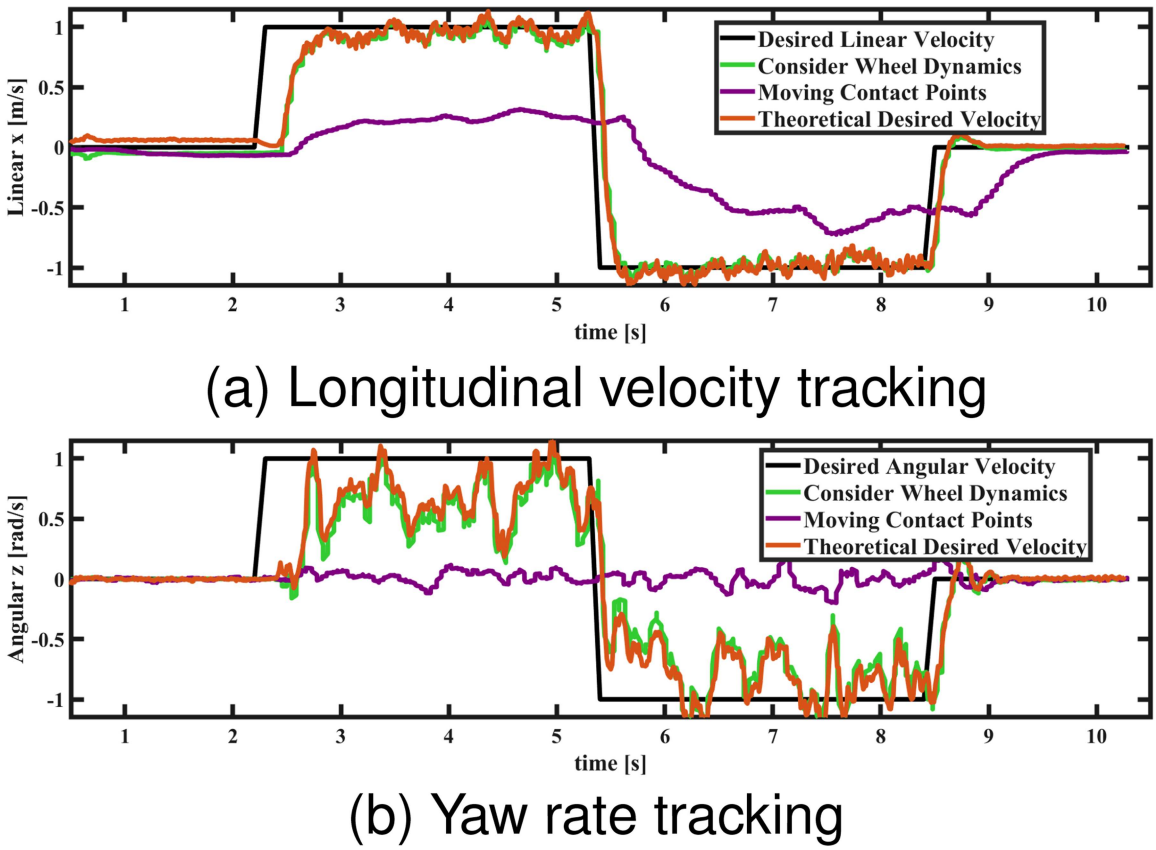


FIGURE 7 | Performance comparison between the proposed wheel-dynamics controller (green) and the moving contact point approximation (purple). Red solid lines denote curves constructed from theoretical equations. (a) Longitudinal velocity tracking (b) Yaw rate tracking.

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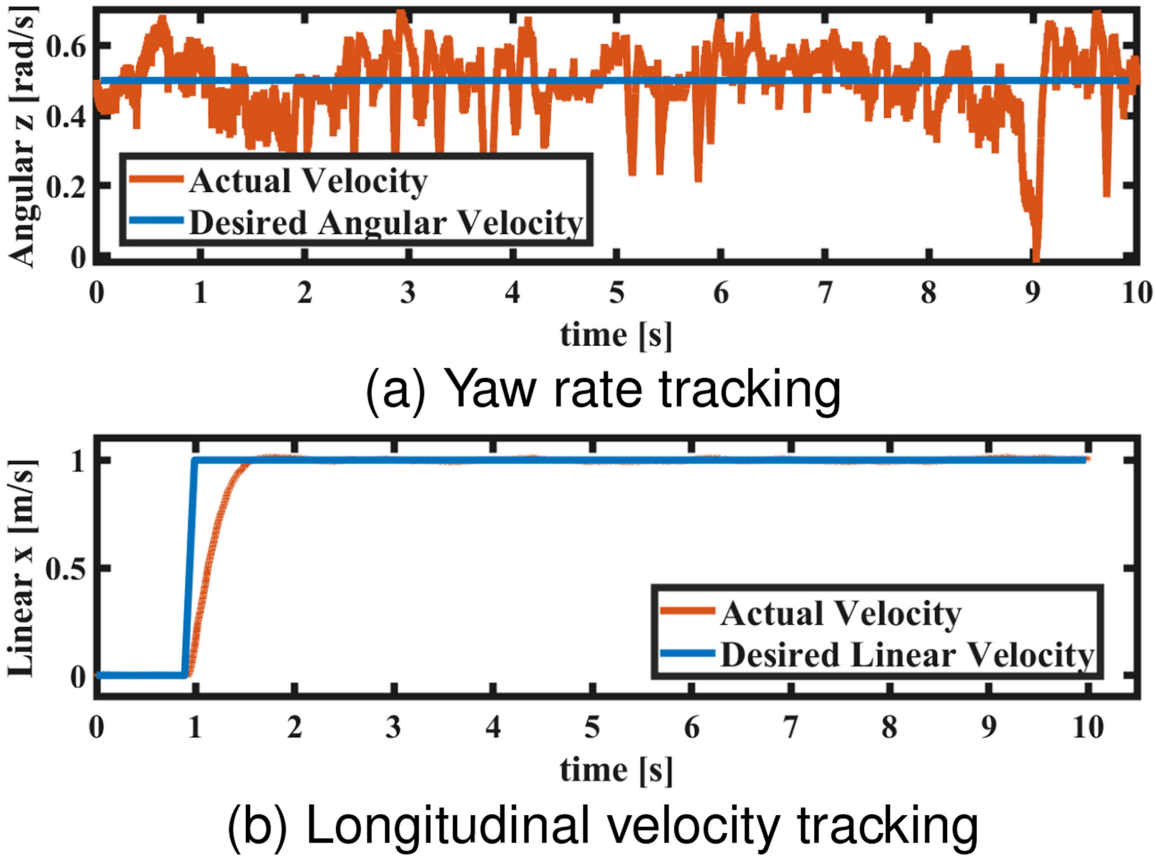


FIGURE 8 | Tracking performance of the proposed framework applied to a traditional wheeled-legged robot with external swing joints in simulation. (a) Angular velocity tracking: The robot achieves yaw rate tracking with an error of 13.25% under reference commands. (b) Linear velocity tracking: The robot achieves linear velocity tracking with an error of 4.12%. Test conditions: simulation environment with ground friction coefficient $\mu = 0.6$, traditional wheeled-legged robot platform with 4 swing joints (one per leg). The results demonstrate the framework's adaptability and transferability to different robot platforms with varying kinematic configurations.

for each controller, and the median trial was selected for performance analysis to ensure robustness against outliers and environmental variations. As demonstrated in Figure 7, both controllers executed identical velocity profiles (first 3 s: 1 m/s longitudinal velocity, 1 rad/s yaw rate; next 3 s: −1 m/s, −1 rad/s) on grassy terrain using the median trial data. The theoretical predictions in Figure 7 show good agreement with experimental measurements, demonstrating the validity of the proposed wheel dynamics models. The baseline MPC exhibited three fundamental shortcomings: (1) 34% slower longitudinal velocity response due to unmodeled tire dynamics; (2) Inability to fully track desired velocities; (3) Complete reliance on external swing joints for steering, making it impossible to achieve steering in skid-steering wheel-legged robots.

The proposed method addresses these issues through: (1) Integration of wheel dynamics models; (2) Coordinated optimization of wheel speed, rotation velocity, and torque; (3) Utilization of skid steering to eliminate dependence on external swing motors. Quantitative results show a 49.86% reduction in trajectory tracking error.

5.6 | Application to Traditional Wheeled-Legged Robots With External Swing Joints

To demonstrate the framework's adaptability and generalization capability, we applied the proposed control algorithm to a traditional wheeled-legged robot platform equipped with external swing joints. Unlike the LegoWheel platform which relies solely on skid-steering, this traditional platform can achieve steering through both differential wheel speeds (skid-steering) and swing joint actuation. The key adaptations required for this extension are as follows:

Skid Steering Implementation: To achieve skid-steering on the traditional platform, the MPC controller was configured to generate differential wheel speed commands while maintaining swing joints in a neutral position. The wheel speed difference was directly controlled to achieve desired yaw rates, following the same ICR-based formulation as the LegoWheel platform. This approach enables the traditional robot to utilize skid-steering as an alternative or complementary steering mechanism to swing joint actuation.

Simulation Validation: The adapted framework was applied to a traditional wheeled-legged robot with four swing joints (one per leg) in a simulation environment with a ground friction coefficient of $\mu = 0.6$. Simulation results demonstrated successful implementation of skid steering: (1) The robot achieved yaw rate tracking with an error of 13.25% and linear velocity tracking with an error of 4.12% (see Figure 8); (2) The framework successfully coordinated wheel differential speeds and swing joint motions when both steering mechanisms were active. These results validate that the proposed framework's

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core principles—switched-system dynamics modeling, hierarchical control architecture, and wheel dynamics integration—are transferable to different robot platforms with varying kinematic configurations.

6 | Conclusion

This paper presents a holistic whole-body control framework for skid-steering wheeled-legged robots, introducing two novel models: a switched-system wheel–terrain interaction dynamics model and a switched-system skid-steering kinematics model. Through extensive experimental validation across six distinct categories of unstructured terrains, the proposed framework demonstrates three significant technological advancements compared to state-of-the-art approaches:

First, the proposed whole-body control framework for skid-steering wheeled-legged robots demonstrates superior tracking performance compared to both model-free reinforcement learning approaches and simplified contact-point models. Second, our hierarchical control architecture enables synergistic coordination between wheel dynamics, contact forces, and motion tracking through prioritized optimization. Third, the integrated skid-steering mechanism maintains precise control while eliminating dependence on external swing joints, as evidenced by: (1) accurate yaw rate tracking, (2) robust response to velocity step changes, and (3) elimination of abnormal leg-lifting behaviors commonly observed in RL-based controllers.

Current limitations include the reliance on offline trajectory planning for complex obstacles and the need for more precise state estimation. Future work will focus on: (1) developing online trajectory planning capabilities to enhance decision-making in complex environments and (2) designing advanced state estimators specifically optimized for wheeled-legged robots to improve estimation accuracy.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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